Designing for Renters' Energy Decisions

Taylor Brooks · BFA Interaction Design capstone · Riverside College of Design · May 2021

Abstract

Renters pay utility bills they cannot predict and rarely control. This project followed 18 renter households through a six-week diary study, mapped when they make energy decisions, and designed a bill-forecast card that shows next month's likely bill and the one habit that moves it most. In a test with 24 participants, correct answers to "what will your next bill be, roughly?" rose from 52% with a standard bill to 87% with the card.

1. Method

Eighteen households in Queens and Brooklyn kept a photo diary of every moment they thought about energy: adjusting a thermostat, reading a bill, arguing about the air conditioner. I followed up with an interview in each home.

2. Findings

Decisions happen before the bill arrives, not after it. Households understood a bill in dollars, not in kilowatt-hours, and trusted a forecast more when it named one cause they could change.

3. Design: the bill-forecast card

A single card shows the likely bill as a range in dollars, the habit that moves it most this month, and what changing that habit would save. I prototyped it in Figma and tested three versions with 24 participants.

4. Results

With the card, 87% of participants estimated their next bill within ten percent, against 52% with a standard bill. Participants also named a specific change they would make in 19 of 24 sessions.

Sample material for the Job Hunter Agent demo. Everyone and every organization named here is fictional.